

Integrated livestock grazing

Spatial scale: farm/vineyard

About this sustainable management strategy

Integrating livestock into vineyard management offers a nature-based approach to controlling vegetation while supporting ecological processes and reducing the need for mechanical or chemical interventions. Sheep are most commonly used because they efficiently manage ground vegetation and recycle nutrients through manure, while poultry such as hens or geese can contribute to weed suppression, insect control, and soil disturbance. Grazing can be integrated during the winter dormancy period or, under suitable vineyard designs and careful management, throughout parts of the growing season. The timing and intensity of grazing should be adapted to vine development, vegetation growth, and local site conditions to maximise ecological benefits while avoiding damage to vines.

Beyond vegetation management, livestock contribute to nutrient cycling, increase soil biological activity, and promote a more diverse ground vegetation. By replacing or reducing mowing, herbicide applications, and other external inputs, integrated grazing supports a more self-regulating vineyard ecosystem while maintaining productive vineyard management.

Key ecosystem services

- Soil health and fertility (*enhanced nutrient cycling, soil biological activity, and organic matter inputs*)
- Natural weed and vegetation control
- Natural pest control (*support for beneficial insects and natural enemies*)
- Habitat provision and biodiversity conservation (*more diverse ground vegetation and associated wildlife*)
- Erosion control (*maintenance of permanent vegetation cover and improved soil stability*)
- Cultural heritage (*preservation of traditional knowledge and historical land-use practices*)

Considerations for successful implementation

Protection of vines during grazing

The risk of damage depends on the growth stage of the vines, animal species, and grazing duration. Young vineyards and periods of active shoot and fruit development require particular attention, while protective measures or temporary exclusion of livestock may be necessary to prevent browsing of shoots, leaves, and grape clusters.

Seasonal grazing management

The timing and duration of grazing should reflect vine phenology, vegetation growth, and soil conditions. Winter grazing generally presents fewer risks, whereas grazing during the growing season requires closer supervision. Rotational grazing can help maintain vegetation cover while avoiding overgrazing and excessive soil disturbance.

Infrastructure and operational planning

Successful grazing requires appropriate fencing, access to drinking water, and, where necessary, shelter for livestock. Grazing activities should also be coordinated with vineyard operations such as pruning, spraying, and harvesting.

Regular monitoring of vineyard conditions

Routine observations of vine condition, vegetation cover, soil moisture, and animal behaviour help identify potential issues such as excessive browsing, soil compaction, or uneven grazing. Monitoring also allows stocking density or grazing duration to be adjusted in response to seasonal conditions.